



Subsonic Mk3

Passive under-desk Subwoofer

Build Guide



You're in the right place

Whether this is your first speaker build or you've made plenty before, follow the steps in order for a clean build and a great result.

Take your time with the details that affect fit and finish.

Build time

One afternoon for most builders, excluding print time.

Skill level

Beginner

The biggest challenge

Sourcing the components

The end result

By the end of this build, you'll have a finished subwoofer you can be proud of, and a proper piece of audio gear you made yourself.





Gather these tools and supplies for the build.

You can organise the remaining tools and supplies while the parts are printing.

Reliable FDM 3D printer

Minimum build volume: W 220 mm × D 250 mm × H 250 mm

Tools you'll need

- Soldering iron and solder
- 3 mm Allen key
- Screwdriver
- Needle-nose pliers
- Printed binding-post hex driver
- Hammer
- Wire cutters and wire strippers
- Scissors (for gasket tape)
- Ruler or measuring tape

What you'll need to supply

2.3 kg of PLA per subwoofer. This design is optimised for PLA. Other materials may work, but PLA is recommended.

These settings have been tested on common FDM printers to produce parts that fit and assemble correctly.

Recommended material

- **PLA** (tested and validated)

Other materials may work, but this build guide assumes PLA.

Part orientation

- Parts are pre-oriented for printing. Do not rotate them.
- Check the orientation against the examples on the next page.

Print time

- The main enclosure may take 48 hours or more to print. This is normal.
- The baffle may print in as little as a few hours.
- Load the parts into your slicer to check the time on your printer.

Print settings

- Nozzle size: 0.4 mm
- Layer height: 0.2 mm
- Line width: 0.6 mm (important for overhangs, print time and strength)
- Perimeters: 3
- Infill: 15% (cubic)
- Top shell thickness: 1.5 mm
- Bottom shell thickness: 1.5 mm
- Slicer supports: Off
- Aux/chamber fan: Off

Use the settings above for a reliable fit.

Slicer supports are not needed. Removable support panels are built into the model.

Common mistakes

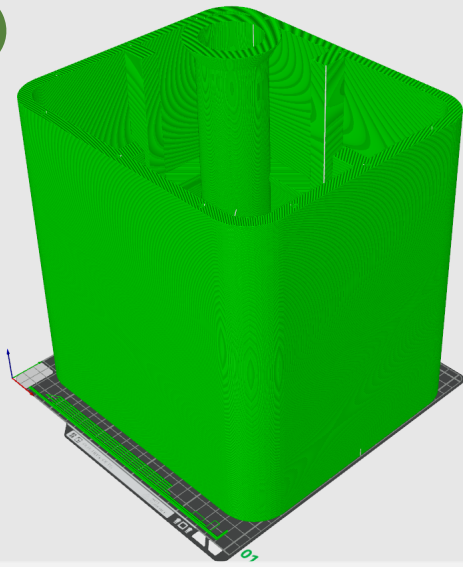
- Increasing infill "just to be safe"
- Adding slicer-generated supports
- Printing in the wrong orientation

Start with the baffle to confirm your settings before moving on to larger parts.

Print all parts exactly as shown below

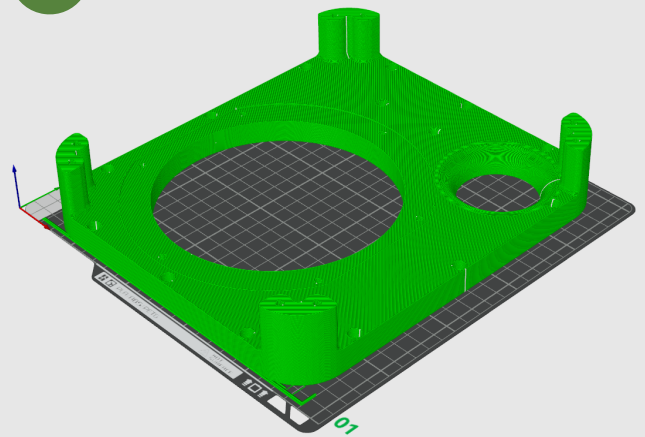
Parts import in the correct orientation by default.

Main enclosure



Print with the flat top surface on the build plate

Front baffle

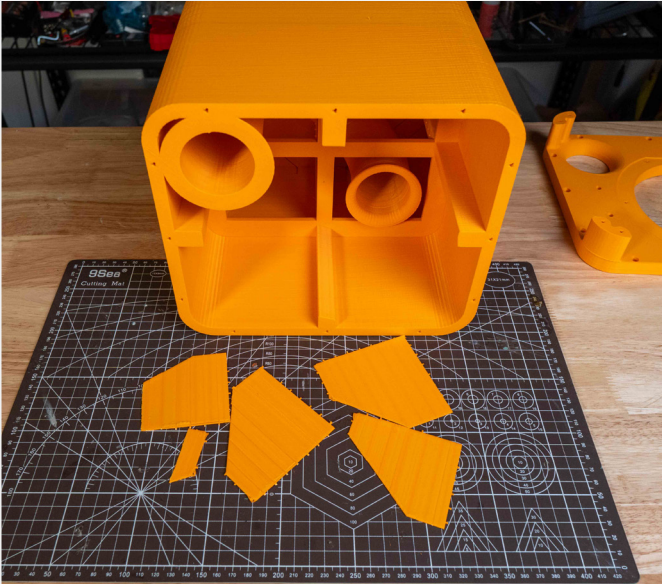


Print with the legs facing up



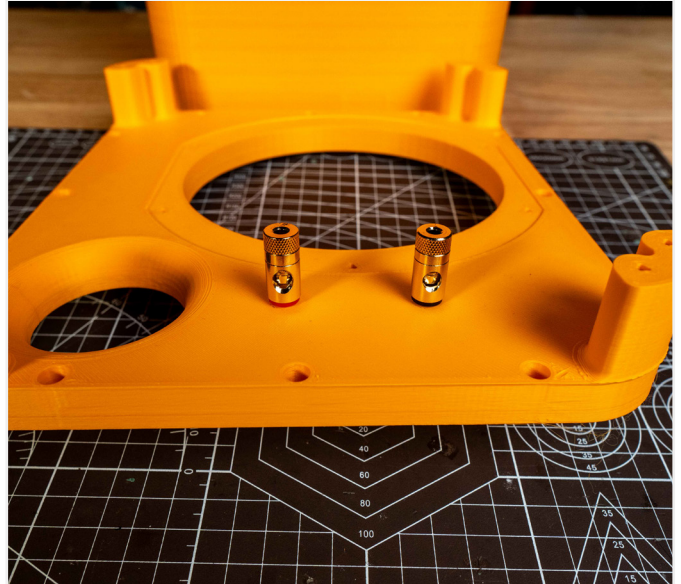
Keep the orientation shown. Rotating the parts may require extra supports and cause print failures.

1A Remove the supports



Remove the built-in support panels from inside the enclosure. If needed, gently knock them out with a screwdriver handle.

1B Fit the binding posts



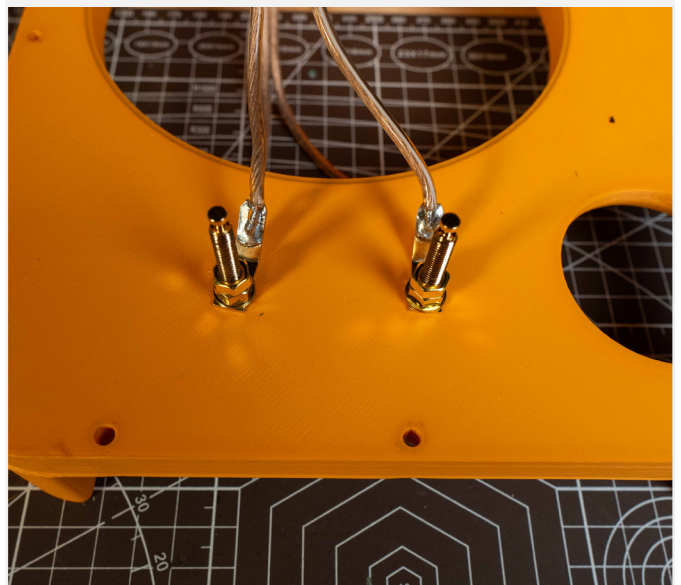
Remove the nuts, washers and solder tabs from the Dayton Audio BPA-38G posts. Gently tap the posts into the holes with a hammer.

1C Solder the tabs



Use 30 cm of speaker wire. Strip the ends, then solder a binding-post tab to each conductor at one end.

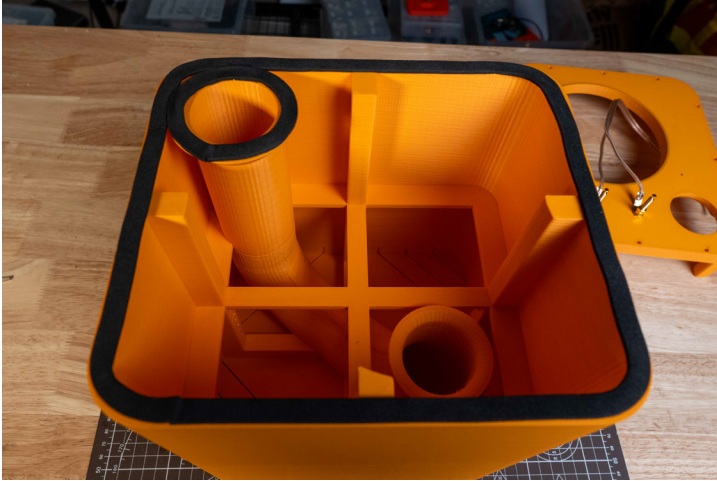
1D Wire up the posts



Refit the solder tabs, washers and nuts. Tighten with the 3D-printed hex driver for the BPA-38G posts, included in the project files.

ASSEMBLE THE ENCLOSURE

2A Apply the gasket tape

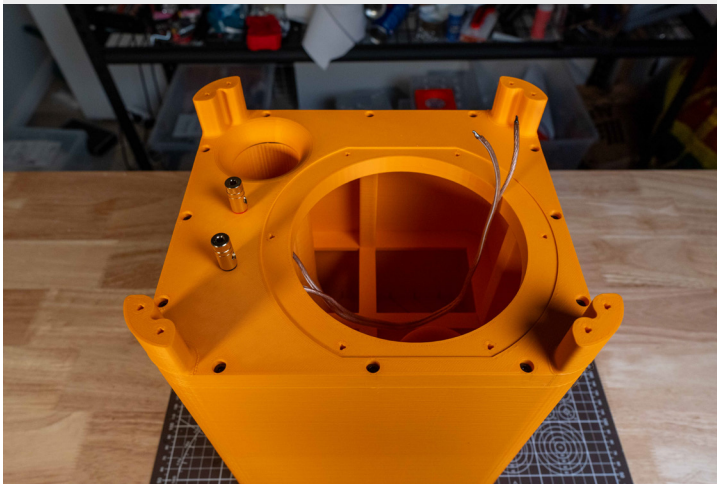


Apply 2–3 mm thick, 10mm wide EVA gasket tape around the full mounting flange, including the port tube. Make a continuous seal.



Use 2–3 mm EVA gasket tape. Thicker tape may require enough force to damage the baffle.

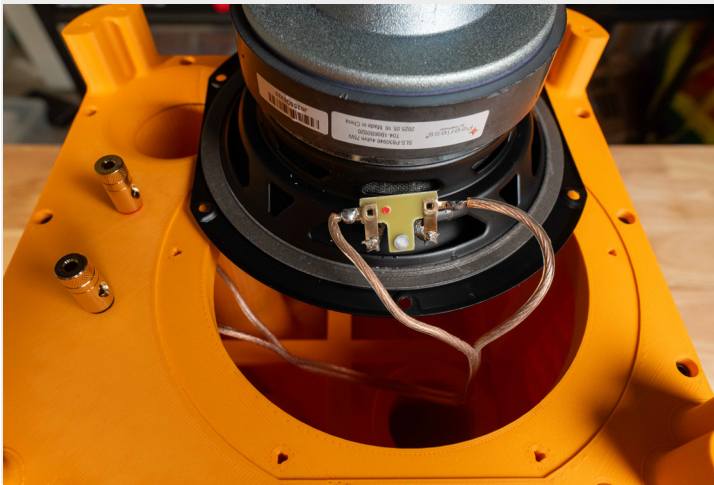
2B Secure the baffle



Secure the baffle with twelve M4 × 20 mm self-tapping screws. Start all screws loosely, then tighten evenly in several passes until the gasket is compressed and the screws are firmly seated.

Next, mount the subwoofer driver.

3A Connect the driver



Solder the loose ends of the speaker wire to the driver terminals. Connect the red binding post to the terminal marked with a red dot, and the black post to the other terminal.

3B Mount the driver



Seat the gasket supplied with the driver around its mounting surface. Secure the driver with six M4 × 12 mm self-tapping screws, tightening evenly in several passes.



Do not overtighten the screws.
Too much force can damage the printed threads.

3C Attach the feet



Attach eight rubber feet, two per printed leg. Use one M4 × 12 mm self-tapping screw for each foot.



Not just assembled. Built.

You printed the parts, assembled the enclosure, mounted the driver and turned a pile of components into a real subwoofer.

That's worth feeling pretty good about.



Help us spread the word

Share a photo of your finished subwoofer on Reddit, Instagram or your favourite DIY audio or 3D printing community.

Your build helps this project reach more builders and might inspire someone else to start their own.

Having trouble? Start here

- **No sound:**
Check the connections at the driver and binding posts.
- **Hissing when bass hits:**
Check the enclosure joints and driver gasket for air leaks.
- **Crackling, buzzing or distortion:**
Check for loose connections, solder joints or objects inside the enclosure.
- **Amplifier shuts down or won't power on:**
Inspect the wiring for short circuits.
- **Weak bass or strange sound:**
Verify polarity at every connection and keep the port unobstructed.

Still stuck? Email help@printyourspeakers.com.
Chat with Hugh and fellow builders on Discord:
<https://discord.gg/SSNYCheZaB>

We value your feedback

Thanks for building with Print Your Speakers. Your feedback helps us improve the kits, guides and build experience.

Fill out the quick survey:
<https://forms.gle/vGdJ8ECs8qqVMPcH6>

Got more to say? Email feedback@printyourspeakers.com.